

DD: FUL-02-STEP-KYC — KYC Two-Level Approval

Developer implementation guide: DD_FUL-02-STEP-KYC-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract DD_FUL-02-STEP-KYC-v1.0.md.

Verdict: ■ New | **Wave:** 1 | **Group III:** Fulfillment | **Version:** v1.0 | **Satellite of:** DD_FUL-02-Order_Capture-v1.0.md Two human-in-the-loop approval steps that gate install dispatch in the KE WIK flow. The customer has paid, but before a contractor is sent out, two Wananchi staff must verify the customer's identity documents. L1 (regional supervisor) reviews first; if approved, Team Leader (operator-level) reviews second. Either rejection terminates the order. Phase: IN_KYC.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/06_fulfillment/DD_FUL-02-STEP-KYC-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- GET /api/groups/kenya-l1-kyc/members?operatorCode=WIK
- GET /engine-rest/history/variable-instance?processInstanceId=...&variableName=kycL1Decision
- POST /api/notify/kyc-approval-needed
- POST /api/orders/ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1/user-tasks/cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3/complete
- POST /api/orders/ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1/user-tasks/cam_task_tl_01HZX9K8M7Q2N4P6R3T5W8Y0Z4/complete
- POST /api/orders/ord_01HZ_REJECT/user-tasks/cam_task_l1_01HZ_REJECT/complete

- POST /api/orders/{orderId}/user-tasks/{taskId}/complete
- POST /engine-rest/task/cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3/complete
- POST /engine-rest/task/cam_task_tl_01HZX9K8M7Q2N4P6R3T5W8Y0Z4/complete

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- create-install-wo
- kenya-kyc-leaders
- kenya-l1-kyc
- kyc-l1-supervisor-approval
- kyc-team-leader-approval
- mark-rejected
- order-create-install-wo
- order-mark-rejected
- uganda-kyc-leaders
- uganda-l1-kyc
- wait-payment

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- PaymentReceived

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.

6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 1 | **Group III:** Fulfillment | **Version:** v1.0 | **Satellite of:** DD_FUL-02-Order_Capture-v1.0.md

Two human-in-the-loop approval steps that gate install dispatch in the KE WIK flow. The customer has paid, but before a contractor is sent out, two Wananchi staff must verify the customer's identity documents. L1 (regional supervisor) reviews first; if approved, Team Leader (operator-level) reviews second. Either rejection terminates the order. Phase: IN_KYC.

1. What this step does

The WIK business policy inherited from the workshop is **no install dispatch without two-level KYC sign-off**. The reason is regulatory + fraud control: ISP licences in Kenya require customer-identity verification before service activation, and contractors are expensive to dispatch — verifying the customer is real and the documents check out reduces wasted dispatches and protects the operator from fraud-driven chargebacks.

The step has two **user tasks** stacked sequentially in the BPMN:

1. `kyc-l1-supervisor-approval` — assigned to candidate group `kenya-l1-kyc`. A regional/franchise supervisor opens the customer's documents (ID front, ID back, address proof, captured by the field agent at order capture time and stored in document storage), checks them against IPRS (national ID registry) where available, and decides APPROVE or REJECT.
2. `kyc-team-leader-approval` — assigned to candidate group `kenya-kyc-leaders`. Only created if L1 approved. The Team Leader sees L1's review as context and decides APPROVE or REJECT independently. This second review is a four-eye control — even if L1 was wrong or compromised, the order doesn't proceed to dispatch without Team Leader sign-off.

A user task is a BPMN step that doesn't run code — it sits in Camunda's task list waiting for a human to claim and complete it through a UI. When Camunda creates the task, it triggers an HTTP connector that POSTs to FUL-02's `NotificationHandlerResource` (FRAMEWORK S3), which dispatches notifications (email, in-app push) to everyone in the task's candidate group. The first person to claim the task in the BO UI sees the documents, makes a decision, completes the task. The BPMN then advances.

The decision outcomes

Stage	Decision	What happens
L1	APPROVE	BPMN creates the Team Leader task; L1's notes carry forward as context.
L1	REJECT	BPMN routes to STEP-VALIDATE's order-mark-rejected worker with reason KYC_REJECTED_L1. Order is terminal.
Team Leader	APPROVE	BPMN advances to STEP-INSTALL. Order is cleared for contractor dispatch.
Team Leader	REJECT	BPMN routes to order-mark-rejected with reason KYC_REJECTED_TEAM_LEADER. Order is terminal. L1's earlier approval doesn't save it — Team Leader has unilateral veto.

What this step deliberately doesn't do

- **Doesn't auto-reject on inactivity.** There's no boundary timer on the user tasks in v1.0. If a supervisor doesn't pick up the task, ops sees the SLA breach on a NOC dashboard (driven by the OrderKycAwaitingL1 Kafka event timestamp) and pages someone. The customer has already paid; auto-cancelling for staff-side delay would be unfair. Future v1.x may add a soft escalation timer that pages without cancelling.
- **KYC two-level is a KE WIK business rule.** Other operators in the architecture may adopt different qualification models via their own BPMN; v1.0 ships only the WIK BPMN with the two-level KYC pattern.
- **Doesn't persist decisions onto customer_order.** Decisions live in Camunda process variables (kycL1Decision, kycL1Actor, kycTeamLeaderDecision, kycTeamLeaderActor, etc.) and are summarized in customer_order_phase_history.metadata when the phase transitions. The only customer_order change is failure_reason + failure_detail on rejection.
- **Doesn't fetch documents from storage itself.** The notification payload carries document references (e.g., idoc_01HX...front); the BO UI fetches actual document images from the document-storage service when the supervisor opens the task.

2. Worked example end-to-end (WIK, both levels APPROVE)

Continuing John Mwangi's order from STEP-PAYMENT. Order ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1, customer cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6, package pkg_INET_V0IP_RES, subscription sub_01HX5M2K9P3Q7R4T6W8Y0Z2N4Q. Payment of 15000 KES landed at 11:23. Camunda just woke wait-payment via PaymentReceived and the BPMN advanced to the L1 user task.

Step 1 — Camunda creates the L1 user task and fires the notification

Camunda creates kyc-l1-supervisor-approval with candidate group kenya-l1-kyc. The HTTP connector fires:

```
POST /api/notify/kyc-approval-needed HTTP/1.1
Host: fulfillment-order-capture.sophix.internal
X-Camunda-Connector-Secret: <shared secret>
Content-Type: application/json

{
  "taskId": "cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3",
  "taskName": "L1 KYC Approval",
  "businessKey": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "orderId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "customerId": "cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6",
  "operatorCode": "WIK",
  "candidateGroup": "kenya-l1-kyc",
  "level": "L1",
  "customerName": "John Mwangi",
  "kycDocuments": [
    { "type": "ID_FRONT", "ref": "idoc_01HX3K9_id_front", "uploadedAt": "2026-05-18T09:00:00Z" },
    { "type": "ID_BACK", "ref": "idoc_01HX3K9_id_back", "uploadedAt": "2026-05-18T09:00:00Z" },
    { "type": "ADDRESS_PROOF", "ref": "idoc_01HX3K9_address_proof", "uploadedAt": "2026-05-18T09:00:00Z" }
  ],
  "deepLink": "/backoffice/orders/ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1/kyc/l1",
  "channels": ["email", "in_app_push"],
  "template": "kyc-approval-l1",
  "urgency": "medium"
}
```

NotificationHandlerResource resolves kenya-l1-kyc membership for WIK via FOUNDATION_AUTH, finds 7 supervisors on shift, writes 7 rows to staff_notification_stub and dispatches email + in-app push.

A task-create event listener also fires the Kafka event:

```
{
  "eventType": "OrderKycAwaitingL1",
  "aggregateId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "timestamp": "2026-05-18T11:23:03.412Z",
  "payload": {
    "orderId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
    "customerId": "cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6",
    "operatorCode": "WIK",
    "userTaskId": "cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3",
    "candidateGroup": "kenya-l1-kyc",
    "createdAt": "2026-05-18T11:23:03.412Z"
  }
}
```

PhaseProjector updates the order:

```
UPDATE customer_order
SET current_phase = 'IN_KYC',
    current_activity_id = 'kyc-l1-supervisor-approval',
    last_updated_at = NOW()
WHERE order_id = 'ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1';
```

Step 2 — L1 supervisor reviews and approves

Supervisor supervisor_user_17 (a regional supervisor for the Nairobi-Karen franchise) opens the BO UI, sees the task in their inbox, clicks through. The UI loads the documents from document storage by their idoc_* references. Supervisor:

- Cross-references John's national ID number against the IPRS lookup endpoint — match.
- Checks address proof (utility bill from KPLC dated last month) — matches HomePass hp_01HX5M2K9P3Q7R4T6W8Y0Z2N4P address.
- Verifies the agent's at-capture metadata (GPS in capturedLocation, agent_01HZX9K8M7Q2N4P6R3T5W8Y0Z1) — geo-coordinates match Karen postal area.

All checks satisfy. Supervisor clicks Approve. BO UI POSTs:

```
POST /api/orders/ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1/user-tasks/cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3/complete HTTP/1.1
Authorization: Bearer <supervisor JWT, role ORDER_BACKOFFICE, groups [kenya-l1-kyc, ...]>
Content-Type: application/json
```

```
{
  "decision": "APPROVE",
  "decisionNotes": "ID verified against IPRS (match); KPLC bill confirms address; agent GPS in correct area",
  "actor": "supervisor_user_17"
}
```

FUL-02's OrderCaptureService:

1. Verifies the actor is in kenya-l1-kyc group for WIK via FOUNDATION_AUTH GET /api/groups/kenya-l1-kyc/members?operatorCode=WIK. ✓
2. Verifies the task is still open. ✓
3. Forwards completion to Camunda:

```
POST /engine-rest/task/cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3/complete HTTP/1.1
```

```
{
  "variables": {
    "kycL1Decision": { "value": "APPROVE",
    "kycL1DecisionNotes": { "value": "ID verified against IPRS (match); KPLC bill confirms address; agent GPS in correct area",
    "kycL1Actor": { "value": "supervisor_user_17",
    "kycL1DecidedAt": { "value": "2026-05-18T11:35:11Z",
  }
}
```

Camunda 204. The BPMN gateway sees kycL1Decision = APPROVE and creates the Team Leader user task.

Step 3 — Camunda creates the Team Leader user task

kyc-team-leader-approval is created with candidate group kenya-kyc-leaders. The HTTP connector fires with an enriched payload that includes L1's review:

```
POST /api/notify/kyc-approval-needed HTTP/1.1
```

```
{
  "taskId": "cam_task_tl_01HZX9K8M7Q2N4P6R3T5W8Y0Z4",
  "taskName": "Team Leader KYC Approval",
  "businessKey": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "orderId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "customerId": "cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6",
  "operatorCode": "WIK",
  "candidateGroup": "kenya-kyc-leaders",
  "level": "TEAM_LEADER",
  "customerName": "John Mwangi",
  "kycDocuments": [ "...same 3 documents as L1..." ],
  "kycL1Decision": "APPROVE",
  "kycL1DecisionNotes": "ID verified against IPRS (match); KPLC bill confirms address; agent GPS in correct area",
  "kycL1Actor": "supervisor_user_17",
  "kycL1DecidedAt": "2026-05-18T11:35:11Z",
  "deepLink": "/backoffice/orders/ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1/kyc/team-leader",
  "channels": ["email", "in_app_push"],
  "template": "kyc-approval-team-leader",
  "urgency": "medium"
}
```

A second Kafka event fires:

```
{
  "eventType": "OrderKycAwaitingTeamLeader",
  "aggregateId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "timestamp": "2026-05-18T11:35:11.512Z",
  "payload": {
    "orderId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
    "customerId": "cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6",
    "operatorCode": "WIK",
    "userTaskId": "cam_task_tl_01HZX9K8M7Q2N4P6R3T5W8Y0Z4",
    "candidateGroup": "kenya-kyc-leaders",
    "level": "TEAM_LEADER",
    "previousLevelActor": "supervisor_user_17",
    "createdAt": "2026-05-18T11:35:11.512Z"
  }
}
```

current_activity_id updates to kyc-team-leader-approval; phase stays IN_KYC.

Step 4 — Team Leader reviews and approves

Team Leader team_leader_user_5 opens the BO UI, sees the task with L1's review visible at the top of the screen. They glance at the documents (UI fetches from doc storage again), see L1's notes, see the IPRS match metadata. They concur.

```
POST /api/orders/ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1/user-tasks/cam_task_tl_01HZX9K8M7Q2N4P6R3T5W8Y0Z4/complete HTTP/1.1
Authorization: Bearer <team leader JWT, role ORDER_BACKOFFICE, groups [kenya-kyc-leaders, ...]>
```

```
{
  "decision": "APPROVE",
  "decisionNotes": "Reviewed L1 decision; concur. Documents in order.",
  "actor": "team_leader_user_5"
}
```

OrderCaptureService verifies group membership, forwards completion to Camunda:

```
POST /engine-rest/task/cam_task_tl_01HZX9K8M7Q2N4P6R3T5W8Y0Z4/complete HTTP/1.1
```

```
{
  "variables": {
    "kycTeamLeaderDecision": { "value": "APPROVE", "type": "String" },
    "kycTeamLeaderDecisionNotes": { "value": "Reviewed L1 decision; concur. Documents in order.", "type": "String" },
    "kycTeamLeaderActor": { "value": "team_leader_user_5", "type": "String" },
    "kycTeamLeaderDecidedAt": { "value": "2026-05-18T11:42:33Z", "type": "String" }
  }
}
```

Camunda 204. BPMN gateway sees kycTeamLeaderDecision = APPROVE and advances to STEP-INSTALL's order-create-install-wo external task. KYC is complete.

Total elapsed in step: ~19 minutes 30 seconds (from L1 task creation at 11:23:03 to Team Leader APPROVE at 11:42:33). Wananchi's SLA target is 30 min combined for both approvals.

3. Worked example — L1 REJECT

Variant of step 2: supervisor finds the address proof is from 18 months ago (Wananchi's KYC policy requires < 3 months). They reject:

```
POST /api/orders/ord_01HZ_REJECT/user-tasks/cam_task_l1_01HZ_REJECT/complete HTTP/1.1
```

```
{
  "decision": "REJECT",
  "decisionNotes": "Address proof dated 2024-11 – older than 3 months policy threshold. Customer must provide updated proof",
  "actor": "supervisor_user_17"
}
```

Camunda gets:

```
{
  "variables": {
    "kycL1Decision": { "value": "REJECT",
    "kycL1DecisionNotes": { "value": "Address proof dated 2024-11 – older than 3 months policy threshold. Customer must provide updated proof",
    "kycL1Actor": { "value": "supervisor_user_17",
  }
}
```

BPMN gateway routes to order-mark-rejected (STEP-VALIDATE's terminal worker) with reason KYC_REJECTED_L1.

That worker writes:

```
UPDATE customer_order
SET current_phase = 'REJECTED',
    failure_reason = 'KYC_REJECTED_L1',
    failure_detail = 'Address proof dated 2024-11 – older than 3 months policy threshold. Customer must provide updated proof',
    last_updated_at = NOW()
WHERE order_id = 'ord_01HZ_REJECT';
```

And emits OrderRejected on Kafka — same event shape as STEP-VALIDATE's rejection event but with failureReason = KYC_REJECTED_L1. DD_NOT-01 consumes and sends the customer:

Sorry, your order could not be processed because your address proof was older than 3 months. Please submit a new order with an updated utility bill.

Note: the customer has already **paid** 15000 KES. The rejection at this point triggers a refund cascade — order-mark-rejected for KYC rejections (specifically KYC_REJECTED_L1 or KYC_REJECTED_TEAM_LEADER with payment_received_amount > 0) emits OrderRefundRequested on top of OrderRejected. Finance processes the refund per the operator's policy. The order itself remains terminally REJECTED — the customer must re-submit if they want service.

4. User tasks

4.1 kyc-l1-supervisor-approval

Field	Value
Purpose	First-level KYC verification by a regional/franchise supervisor. Checks customer documents against ID registry + address proof.
Candidate group	kenya-l1-kyc (operator-scoped; FOUNDATION_AUTH)
Notification endpoint	POST /api/notify/kyc-approval-needed with level = L1
Completion endpoint	POST /api/orders/{orderId}/user-tasks/{taskId}/complete (FUL-02 wrapper; verifies actor's group membership)
Decision values	APPROVE / REJECT
Process variables set on completion	kycL1Decision, kycL1DecisionNotes, kycL1Actor, kycL1DecidedAt
BPMN route on APPROVE	Creates kyc-team-leader-approval user task
BPMN route on REJECT	Routes to STEP-VALIDATE's order-mark-rejected with failureReason = KYC_REJECTED_L1
Boundary timer	None in v1.0 (see R-FUL-02-KYC-6)

4.2 kyc-team-leader-approval

Field	Value
Purpose	Final-level KYC approval by a Wananchi Team Leader. Independent review with L1's decision visible as context. Mandatory before install dispatch.
Candidate group	kenya-kyc-leaders (operator-scoped; FOUNDATION_AUTH)
Notification endpoint	POST /api/notify/kyc-approval-needed with level = TEAM_LEADER + L1's review carried in the payload

Field	Value
Completion endpoint	Same as L1
Decision values	APPROVE / REJECT
Process variables set on completion	kycTeamLeaderDecision, kycTeamLeaderDecisionNotes, kycTeamLeaderActor, kycTeamLeaderDecidedAt
BPMN route on APPROVE	Advances to STEP-INSTALL's order-create-install-wo
BPMN route on REJECT	Routes to order-mark-rejected with failureReason = KYC_REJECTED_TEAM_LEADER
Boundary timer	None in v1.0

5. DB tables touched

customer_order (FRAMEWORK §5.1) — UPDATE

Column	When set	Set by	Example value
current_phase	On BPMN entry to first KYC task	PhaseProjector	IN_KYC
current_activity_id	On each Camunda transition	PhaseProjector	kyc-l1-supervisor-approval → kyc-team-leader-approval → optionally mark-rejected
failure_reason (on REJECT only)	order-mark-rejected	KYC_REJECTED_L1 or KYC_REJECTED_TEAM_LEADER	
failure_detail (on REJECT only)	order-mark-rejected	Decision notes from the rejecting actor	
last_updated_at	On every UPDATE	trigger / PhaseProjector	2026-05-18 11:42:33.512

Sample data — five orders at various KYC stages:

order_id	operator_code	payment_received_amount	current_phase	current_activity_id	failure_reason
ord_01HZ...A	WIK	15000.00	IN_KYC (L1 approved; awaiting Team Leader)	kyc-team-leader-approval	NULL
ord_01HZ...B	WIK	15000.00	IN_KYC (awaiting L1)	kyc-l1-supervisor-approval	NULL
ord_01HZ...C	WIK	15000.00	READY_FOR_INSTALL (Team Leader approved; advanced to STEP-INSTALL)	create-install-wo	NULL
ord_01HZ...D	WIK	15000.00	REJECTED (L1 rejected; refund pending)	end	KYC_REJECTED_L1
ord_01HZ...E	WIK	15000.00	REJECTED (Team Leader rejected after L1 approval; refund pending)	end	KYC_REJECTED_TEAM_LEADER

customer_order_phase_history (FRAMEWORK §5.2)

Multiple rows across the step. For the happy path (both APPROVE):

id	previous_phase	new_phase	new_activity_id	triggered_by	metadata
ordh..._k1	AWAITING_PAYMENT	IN_KYC	kyc-l1-supervisor-approval	message_correlation	{"correlatedMessage": "PaymentReceived", "paymentId": "pmt_01HZ_MPESA_NK4XX7QP9"}
ordh..._k2	IN_KYC	IN_KYC	kyc-team-leader-approval	user_task_completed	{"kycL1Decision": "APPROVE", "kycL1Actor": "supervisor_user_17", "kycL1DecisionNotes": "ID verified against IPRS..."}

id	previous_phase	new_phase	new_activity_id	triggered_by	metadata
ordh..._k3	IN_KYC	READY_FOR_INSTALL	create-install-wo	user_task_complete d	{"kycTeamLeaderDecision":"APPROVE", "kycTeamLeaderActor":"team_leader_user_5", "kycTeamLeaderDecisionNotes":"Reviewed L1 decision; concurr..."}

For L1 REJECT:

id	previous_phase	new_phase	new_activity_id	triggered_by	metadata
ordh..._kr1	IN_KYC	REJECTED	mark-rejected	user_task_complete d	{"kycL1Decision":"REJECT", "kycL1Actor":"supervisor_user_17", "failureReason":"KYC_REJECTED_L1", "decisionNotes":"Address proof dated 2024-11..."}

staff_notification_stub (FRAMEWORK §6.3)

Two notification batches: one when L1 task is created (7 supervisors notified), one when Team Leader task is created (3 leaders notified). Sample for L1:

notification_id	order_id	task_id	candidate_group	dispatch_status	created_at
notif_01HZ_l1_01	ord_01HZ...A	cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3	kenya-l1-kyc	DISPATCHED	2026-05-18 11:23:03.512
notif_01HZ_l1_02	ord_01HZ...A	cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3	kenya-l1-kyc	DISPATCHED	2026-05-18 11:23:03.518
...	...	...	...	...	...

6. Kafka events emitted

sophix.orders.kyc-awaiting-l1

```
{
  "eventType": "OrderKycAwaitingL1",
  "aggregateId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "timestamp": "2026-05-18T11:23:03.412Z",
  "payload": {
    "orderId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
    "customerId": "cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6",
    "operatorCode": "WIK",
    "userTaskId": "cam_task_l1_01HZX9K8M7Q2N4P6R3T5W8Y0Z3",
    "candidateGroup": "kenya-l1-kyc",
    "createdAt": "2026-05-18T11:23:03.412Z"
  }
}
```

sophix.orders.kyc-awaiting-team-leader

```
{
  "eventType": "OrderKycAwaitingTeamLeader",
  "aggregateId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
  "timestamp": "2026-05-18T11:35:11.512Z",
  "payload": {
    "orderId": "ord_01HZX9K8M7Q2N4P6R3T5W8Y0Z1",
    "customerId": "cust_01HX3K9M2P5Q8R1T4W7Y0Z3N6",
    "operatorCode": "WIK",
    "userTaskId": "cam_task_tl_01HZX9K8M7Q2N4P6R3T5W8Y0Z4",
    "candidateGroup": "kenya-kyc-leaders",
    "level": "TEAM_LEADER",
    "previousLevelActor": "supervisor_user_17",
    "createdAt": "2026-05-18T11:35:11.512Z"
  }
}
```

Consumers:

Consumer	Purpose
NOC	SLA monitoring — both <code>OrderKycAwaiting*</code> events tracked; tasks unclaimed >15 min trigger pager alerts (since the customer has already paid, KYC delays directly impact customer experience).
CVM	Funnel KYC stages — drop-offs at L1 vs Team Leader reveal whether documentation policy is stricter at one level than the other.

DD_NOT-01 does NOT consume `OrderKycAwaiting*` — the customer has nothing to do during KYC; no automated message fires. The customer's next notification is when KYC completes and install is dispatched (STEP-INSTALL's `OrderInstallDispatched` → DD_NOT-01 SMSes the install date).

7. Step-pinned rules

ID	Rule
R-FUL-02-KYC-1	Two-level approval mandatory in WIK before install dispatch: L1 supervisor (kenya-l1-kyc) → Team Leader (kenya-kyc-leaders). Both must APPROVE. Team Leader is created only if L1 approved; if L1 rejected, Team Leader never sees the task.
R-FUL-02-KYC-2	Either level rejecting → terminal REJECTED state via <code>order-mark-rejected</code> (STEP-VALIDATE's terminal worker, reused). No appeal path in v1.0 — customer submits a new order if circumstances change. Rejection with <code>payment_received_amount > 0</code> triggers <code>OrderRefundRequested</code> via <code>order-mark-rejected</code> (refund cascade).
R-FUL-02-KYC-3	L1's decision (<code>kycL1Decision</code> , <code>kycL1DecisionNotes</code> , <code>kycL1Actor</code>) is passed forward as context fields in the Team Leader's notification payload. Team Leader is not blind — they see what L1 thought.
R-FUL-02-KYC-4	Completion endpoint verifies the actor is in the task's candidate group via <code>FOUNDATION_AUTH</code> before forwarding to Camunda. Out-of-group attempts → 403. The actor's user ID must also match the JWT subject — preventing one person from "approving on behalf of" another.
R-FUL-02-KYC-5	Candidate groups are operator-scoped in <code>FOUNDATION_AUTH</code> . v1.0 carries only <code>kenya-l1-kyc</code> and <code>kenya-kyc-leaders</code> . Future WUG/WTZ deployments adopting KYC two-level would have their own candidate groups (e.g., <code>uganda-l1-kyc</code> , <code>uganda-kyc-leaders</code>). Group naming convention: <code>{country}-{level}-kyc</code> and <code>{country}-kyc-leaders</code> .
R-FUL-02-KYC-6	NO boundary timer on KYC user tasks in v1.0. KYC approvals are not auto-rejected for inactivity (would be unfair to the customer who already paid). Ops monitors via NOC SLA dashboards driven by <code>OrderKycAwaiting*</code> Kafka event timestamps; tasks unclaimed >15 min trigger a pager alert.
R-FUL-02-KYC-7	The KE WIK BPMN owns these two user tasks. Other operator BPMNs in the architecture may use different qualification models; v1.0 ships only the WIK BPMN.
R-FUL-02-KYC-8	Decisions are NOT persisted as dedicated <code>customer_order</code> columns. They live in Camunda's history (<code>GET /engine-rest/history/variable-instance?processInstanceId=...&variableName=kycL1Decision</code>) and <code>customer_order_phase_history.metadata</code> . Downstream needing filterable orders by KYC outcome should use the <code>OrderRejected</code> Kafka event (<code>failureReason</code> discriminates).
R-FUL-02-KYC-9	Document references in the notification payload are opaque IDs (<code>idoc_*</code>); the BO UI fetches actual document images from the document-storage service when the supervisor opens the task. This step's payload deliberately doesn't embed image bytes — would bloat the notification flow and create stale-image risk.